DATA ANALYSIS REPORT

Ames Housing

Neighborhood sale prices and property characteristics

Prepared 2026-10-18 | Ames, Iowa home sales

THE QUESTION

How do sale prices vary across Ames neighborhoods, and are higher-priced neighborhoods also characterized by newer homes, larger lots, or higher overall quality?

KEY FINDINGS

$319,000
Highest neighborhood median
StoneBr; 51 sales

$88,250
Lowest neighborhood median
MeadowV; 37 sales

0.877
Price and quality rank correlation
Strongest observed association

OVERALL ANSWER

Yes. Higher-priced neighborhoods generally had higher average quality, newer homes, and larger lots. The pattern was clearest for quality; age and lot area were less consistent. These are neighborhood-level associations, not proof that a feature caused a home's sale price.

Data and method

The source file, AmesHousing-1.csv, contains 2,930 home sales and 82 fields. Each record describes a sold home in Ames, Iowa. The analysis groups homes into 28 neighborhoods and focuses on sale price, neighborhood, year built, lot area, and overall quality.

Cleaning decisions

- The original CSV was preserved; no duplicate rows or property IDs were found.
- Whitespace was trimmed and numeric fields were checked and normalized.
- Source codes for absent features were standardized; structurally absent basement or garage measurements were set to zero only when absence was clear.
- Unknown measurements stayed blank. All 2,930 records had values for the five analysis fields, so none were excluded.

Measures and interpretation

Median sale price is the middle sale price in a neighborhood, less affected by unusually high or low transactions than the mean. Average year built, lot area, and quality describe neighborhood characteristics. Spearman's rank correlation (rho) measures whether neighborhoods ranking higher on one measure tend to rank higher on another; it does not measure causation.

Code excerpts from the reproducible workflow

```
for neighborhood, homes in groups.items():
    median_price = median(int(h['SalePrice']) for h in homes)
    avg_year = mean(int(h['Year Built']) for h in homes)
    avg_lot = mean(int(h['Lot Area']) for h in homes)
    avg_quality = mean(int(h['Overall Qual']) for h in homes)
```

Rank association is calculated across the 28 neighborhood summaries:

```
rho_quality = spearman(median_prices, average_quality)
rho_year = spearman(median_prices, average_year_built)
rho_lot = spearman(median_prices, average_lot_area)
```

Neighborhood data at a glance

Below are the three highest- and lowest-median-price neighborhoods. The complete 28-neighborhood comparison follows in the summary table artifact.

Neighborhood	Sales	Median price	Avg. year	Avg. lot area	Avg. quality
StoneBr	51	$319,000	1999.4	11,383	8.37
NridgHt	166	$317,750	2005.6	11,217	8.28
NoRidge	71	$302,000	1995.4	12,976	7.93
IDOTRR	93	$106,500	1927.7	8,826	4.73
BrDale	30	$106,000	1971.8	1,841	5.73
MeadowV	37	$88,250	1972.2	2,109	4.38

Neighborhood-level Spearman rank correlations

Average overall quality	**0.877**	Strongest positive alignment
Average year built	**0.681**	Positive alignment
Average lot area	**0.570**	Positive, less consistent alignment

Reading the results carefully

A positive rho means neighborhoods with higher ranks on one measure tended to have higher ranks on the other. Neighborhood means can hide differences between individual homes. Landmrk has only 1 sale and GrnHill 2, so their neighborhood estimates are especially uncertain. The associations do not establish cause and effect.

Summary table

Neighborhood summaries sorted from highest to lowest median sale price. The table includes sales count and average year built, lot area, and overall quality.

Ames Housing by Neighborhood

Ranked by median sale price · summary across 2,930 sales

Neighborhood	Number of sales	Median sale price	Average year built	Average lot area (sq. ft.)	Average overall quality
StoneBr	51	$319,000	1999.4	11,383	8.4
NridgHt	166	$317,750	2005.6	11,217	8.3
NoRidge	71	$302,000	1995.4	12,976	7.9
GrnHill	2	$280,000	1992.0	9,001	7.0
Veenker	24	$250,250	1981.7	15,482	7.2
Timber	72	$232,106	1995.3	18,234	7.2
Somerst	182	$225,500	2004.6	7,611	7.4
Crawfor	103	$200,624	1946.8	11,650	6.3
CollgCr	267	$200,000	1997.9	10,000	6.7
Greens	8	$198,000	1979.4	3,884	8.0
ClearCr	44	$197,500	1966.8	24,842	6.0
Blmngtn	28	$191,500	2005.0	3,399	7.1
Gilbert	165	$183,000	1998.3	11,342	6.5
NWAmes	131	$181,000	1975.6	11,662	6.3
SawyerW	125	$180,000	1990.2	10,155	6.3
Mitchel	114	$153,500	1981.9	12,020	5.5
NPkVill	23	$143,750	1976.3	2,756	6.3
NAmes	443	$140,000	1959.9	10,040	5.3
Landmrk	1	$137,000	1993.0	3,612	6.0
SWISU	48	$136,200	1930.4	7,615	5.4
Sawyer	151	$135,000	1962.9	10,372	5.1
Blueste	10	$130,500	1982.6	2,346	6.6
BrkSide	108	$126,750	1932.4	6,960	5.1
Edwards	194	$125,000	1956.9	10,356	5.1
OldTown	239	$119,900	1922.3	8,241	5.2
IDOTRR	93	$106,500	1927.7	8,826	4.7
BrDale	30	$106,000	1971.8	1,841	5.7
MeadowV	37	$88,250	1972.2	2,109	4.4

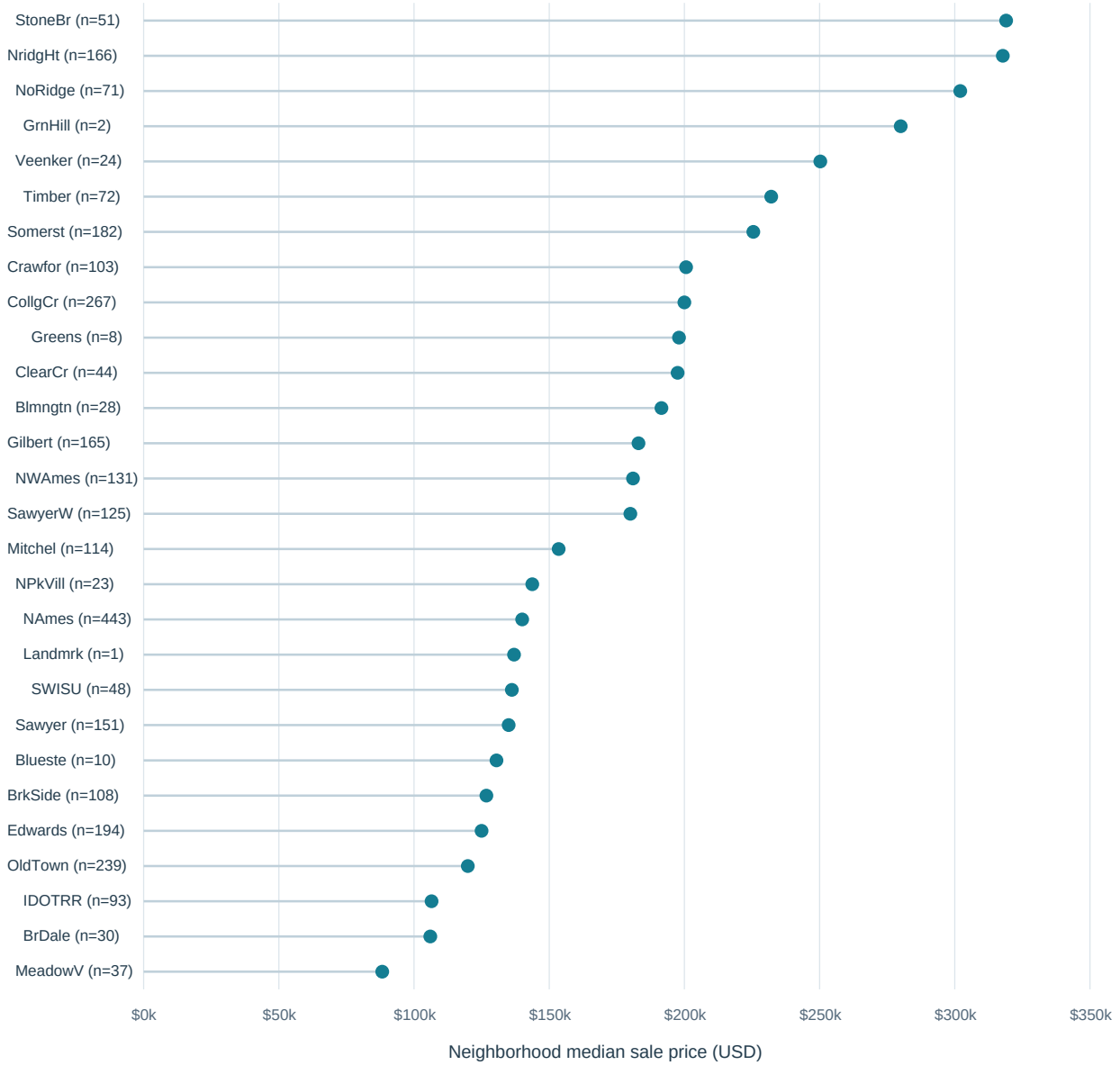
Source: cleaned Ames housing dataset; 2,930 sales in 28 neighborhoods.

Visualization 1 - median sale price

The dot plot ranks neighborhoods by median sale price. The count beside each label shows how many sales contribute to that neighborhood summary.

Typical sale price by neighborhood

Median sale price; sales count shown beside each neighborhood



Median is the neighborhood midpoint sale price.

Visualization 2 - price and overall quality

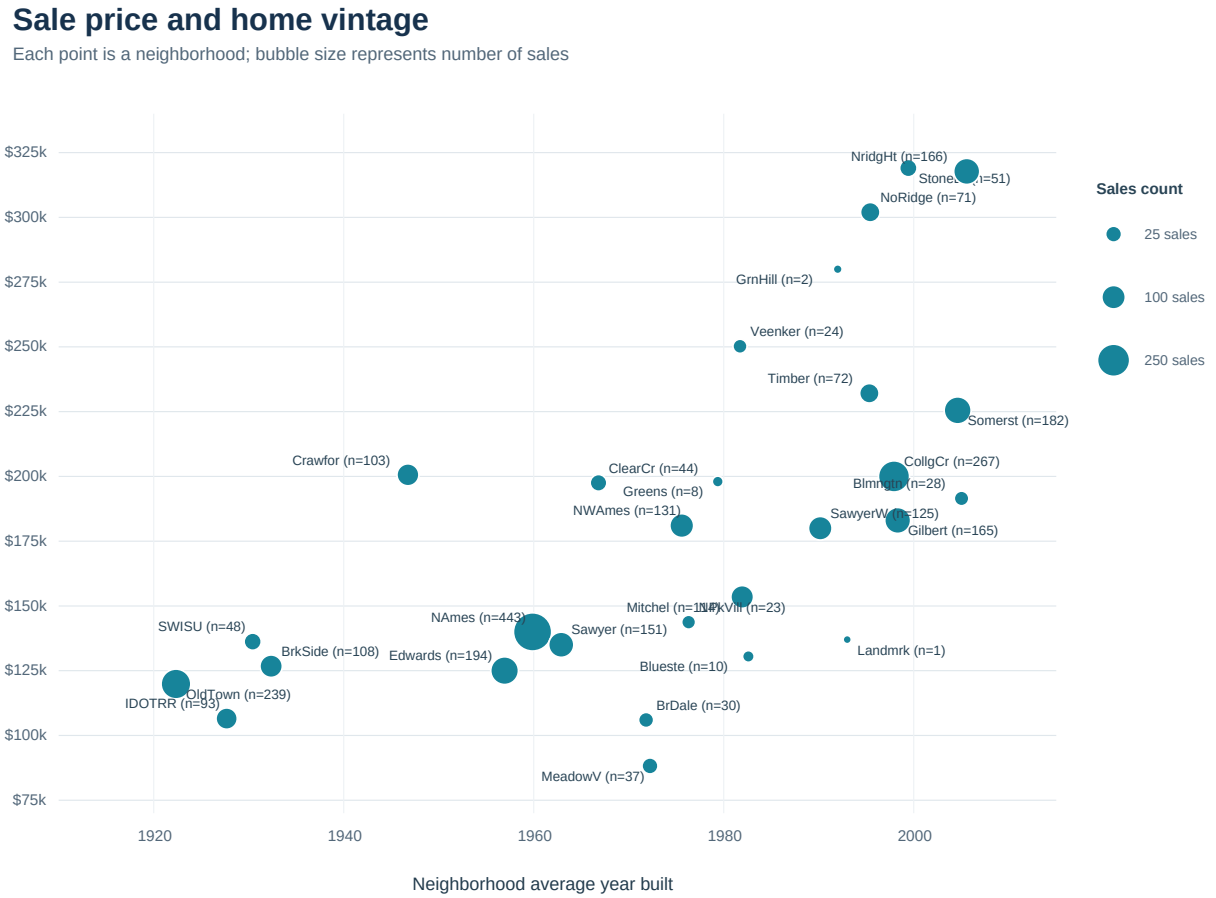
Each bubble represents a neighborhood. Higher points indicate higher median prices; bubble size represents sales count. Quality has the strongest rank association with price (rho = 0.877).



Overall quality is averaged within each neighborhood.

Visualization 3 - price and year built

This chart compares median sale price with average year built. A later average year means the neighborhood's homes tend to be newer. The rank association is positive (rho = 0.681).



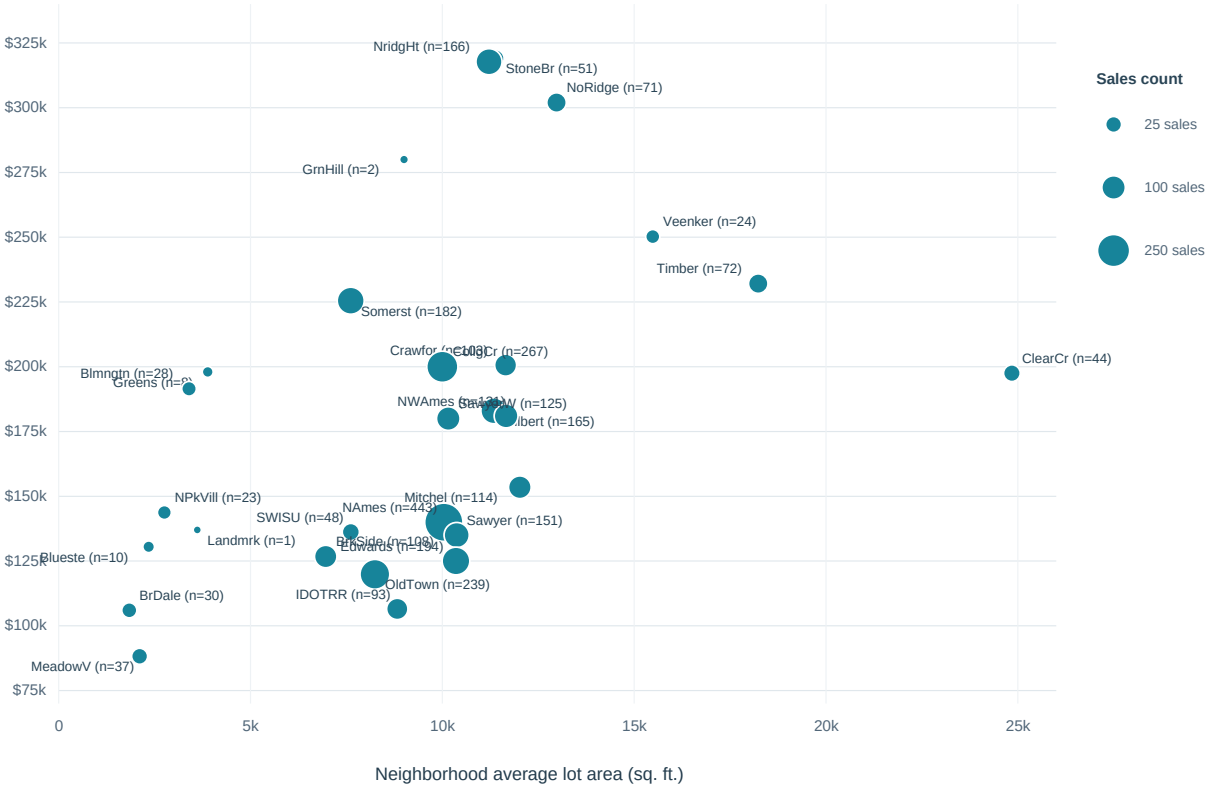
Average year built is a neighborhood summary, not an individual home age estimate.

Visualization 4 - price and lot area

This chart compares median sale price with average lot area. The positive rank association is more moderate (rho = 0.570), and large average lots do not always correspond to the highest median prices.

Sale price and lot area

Each point is a neighborhood; bubble size represents number of sales



Lot area is measured in square feet.

Visualization 5 - neighborhood rank heatmap

Each cell shows a neighborhood's rank for that measure; rank 1 is highest. Green cells indicate higher ranks and warmer colors indicate lower ranks. Rows are ordered by median sale price.

Neighborhood rank profile

Rank 1 is highest within each measure; cell text shows rank

	Median price	Year built	Lot area	Overall quality
StoneBr (n=51)	1	4	8	1
NridgHt (n=166)	2	1	10	2
NoRidge (n=71)	3	7	4	4
GrnHill (n=2)	4	10	16	9
Veenker (n=24)	5	14	3	6
Timber (n=72)	6	8	2	7
Somerst (n=182)	7	3	20	5
Crawfor (n=103)	8	24	7	15
CollgCr (n=267)	9	6	15	10
Greens (n=8)	10	15	22	3
ClearCr (n=44)	11	20	1	18
Blmngtn (n=28)	12	2	24	8
Gilbert (n=165)	13	5	9	12
NWAmes (n=131)	14	17	6	13
SawyerW (n=125)	15	11	13	14
Mitchel (n=114)	16	13	5	20
NPkVill (n=23)	17	16	25	16
NAmes (n=443)	18	22	14	22
Landmrk (n=1)	19	9	23	17
SWISU (n=48)	20	26	19	21
Sawyer (n=151)	21	21	11	25
Blueste (n=10)	22	12	26	11
BrkSide (n=108)	23	25	21	24
Edwards (n=194)	24	23	12	26
OldTown (n=239)	25	28	18	23
IDOTRR (n=93)	26	27	17	27
BrDale (n=30)	27	19	28	19
MeadowV (n=37)	28	18	27	28

Higher rank — Lower rank

Rank colors support comparison; the printed rank in each cell is the precise value.

Narrative summary

Typical sale prices differed widely across Ames neighborhoods. The highest median sale price was in StoneBr at $319,000, based on 51 sales. NridgHt was close behind at $317,750 across 166 sales, followed by NoRidge at $302,000 across 71 sales. At the other end, MeadowV had the lowest median at $88,250 across 37 sales.

A median is the midpoint sale price: half the sales in that neighborhood were higher and half were lower. It helps describe a typical sale without being pulled as strongly by a few unusually expensive homes.

Across neighborhoods, sale price tended to rise along with average overall quality, average year built, and average lot area. To measure these patterns, I used Spearman rank correlation. It compares how neighborhoods rank on two measures: a positive value means neighborhoods ranked higher on one measure also tended to rank higher on the other. The relationship was strongest for overall quality (rho = 0.877), followed by year built (rho = 0.681) and lot area (rho = 0.570). In plain terms, neighborhood median prices aligned most closely with average quality, and somewhat less consistently with newer homes and larger lots.

These are neighborhood-level patterns in this dataset; they do not show that home age, lot size, or quality caused sale prices to be higher. Neighborhoods with very few sales, such as Landmrk (1 sale) and GrnHill (2 sales), should also be interpreted cautiously because their median prices and averages are based on limited observations.

Overall, the answer is yes: higher-priced neighborhoods generally had newer homes, larger lots, and higher overall quality. The connection was clearest for quality; the relationships with home age and lot area were more moderate and had exceptions. These patterns describe how neighborhoods compared in this dataset, rather than predicting the price of an individual home.